



Buildup and relaxation of stress in chemically strengthened glass



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ABSTRACT

Cooper–Krohn analysis of stress generation in soda lime silicate (“SLS”) glasses during ion exchange strengthening suggests that the glass should develop 2.5 to 3 GPa surface compression (biaxial stress), though only ~600 MPa values are observed experimentally. In this paper, we suggest that this “network dilation anomaly” is largely resolved. MD simulations were carried out to monitor volume and pressure changes after ion-swap. Surface compression in an SLS glass ion-exchanged as a function of temperature and time in KNO₃ bath was measured to follow the viscous relaxation. We show that biaxial surface compression builds initially to the expected ~2.5 GPa; thereafter fast secondary structural (“β”) relaxations, whose time constants span picoseconds to nanoseconds, cause the surface compression to relax rapidly to ~1.5 GPa at the ion exchange temperature. Stress relaxes thereafter due to “α” primary structural relaxations associated with viscous flow. Recovery occurs to a transition state which is somewhat short of the expected glass volume. Full volume is recovered at higher temperatures due to increased flow.

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1. Introduction

Since the discovery of the glass chemical strengthening process in the early 1960s, the single most important commercial application has been the aircraft cockpit windshields which constitute two or three layers of glass laminated with polyvinyl butyrates or polyurethane to withstand bird hits at speeds of 400 knots per hour. The primary advantage of the chemical strengthening process over the traditional thermal tempering process is the development of high surface compression, of the order of 600 MPa, which helps increase the windshield modulus of rupture immensely. In recent years, the use of scratch-resistant strengthened thin cover glasses in personal mobile electronic devices such as cell phones and electronic tablets has renewed interest in the chemical strengthening process. Other applications such as chemically strengthened glass cartridges for autoinjectors as an emergency antidote to avoid anaphylactic shock in severe cases of allergies to bee stings, peanuts and shellfish have also drawn attention because of their importance in lifesaving situations. Benefits and drawbacks of the chemical strengthening process relative to the thermal tempering process can be found elsewhere [1].

The development of surface compression in metals using nitridation was described by analogy to thermal stresses by Richmond et al. [2]. This formalism was then adopted to describe ion exchange strengthening

in alkali-containing glass by Cooper and Krohn [3]. For a semi-infinite sodium-containing glass sheet of thickness $2L$ which is exchanged symmetrically in the x -direction (thickness direction) by immersing in a molten salt such as KNO₃, the stresses at location x are given by [1]:

$$[\sigma_{yy}]_x = [\sigma_{zz}]_x = -BEC/(1-\nu) + [BE/2L(1-\nu)] \int_0^{2L} Cdx \quad (1)$$

and $[\sigma_{xx}]_x = 0$,

where E = Young's modulus, ν = Poisson's ratio, and B = linear network dilation coefficient “LNDC”

$$= 1/3[\partial \ln V / \partial C] \quad (2)$$

The first term represents a compression due to stuffing from the local concentration of the invading ion and the second term represents a balancing tension due to elastic compatibility. Cooper also proposed that an unconstrained K-exchanged glass will have a molar volume V_{mk} which corresponds to that of the compositionally equivalent as-melted (“CEAM”) glass. Hence, the LNDC should be given by

$$B = [1/3C_0] \ln(V_{mk}/V_{mNa}) \approx [V_{mk} - V_{mNa}]/3V_{mNa}C_0 \quad (3)$$

where V_{mNa} is the molar volume of the parent Na-containing glass and C_0 is the mol% or mol fraction of alkali in glass that exchanges. The factor of 3 converts a volume coefficient to a linear coefficient.

The quantity BC_0 represents the linear expansion of the glass network which is constrained by the unexchanged substrate from

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expanding to its CEAM value and is responsible for the generation of surface compression.

$$[\sigma_{yy}]_0 = [\sigma_{zz}]_0 \approx -BEC_0/(1-\nu) = -[E/(1-\nu)][V_{mK}-V_{mNa}]/3V_{mNa} \quad (4)$$

Stress measurements in a semi-infinite ion-exchanged plate glass specimen are often made using two techniques:

- (1) using ellipsometry (or refractometry) on the flat face of the chemically strengthened x-plane nondestructively, and
- (2) using optical birefringence through a thin slice containing the diffusion direction x (see ASTM C-1422, Ref [5]).

In the first experiment, one measures the biaxial stress $[\sigma_{yy}]_0$ which is the planar stress on the surface assuming plane-strain condition. In the second experiment, one measures $\sigma_y - \sigma_x$ which is exactly equal to $[\sigma_y]_0$ at the surface under a plane-stress assumption. From the measured $[\sigma_y]_0$, it is customary to report the “unbroken plate” stress which is essentially the biaxial stress $[\sigma_{yy}]_0$ by using the relationship $\sigma_y/(1-\nu) = \sigma_{yy}$.

It has long been recognized [4] that values of B computed from measured molar volumes of the host glass and that of the CEAM glass are usually 3 to 5 times those computed from the experimentally measured stresses. The sizable discrepancy of surface compression developed during ion exchange between that measured and that expected from theory, otherwise called the “dilation anomaly”, has remained controversial for nearly four decades.

Measured values of $[\sigma_y]_0$ or $[\sigma_{yy}]_0$ in chemically strengthened soda lime silicate glasses are usually only about 400 to 600 MPa – far lower than that calculated from as-melted molar volumes (~2 to 2.5 GPa). Sane and Cooper [6] were unable to relate the disparity to viscous flow during ion exchange. Garfinkel and King [7], Varshneya [8], and Glebov et al. [9] suggested that some initial network accommodation of the invading ion, a “plastic deformation”, is responsible for reduced buildup of stress. In particular, Varshneya [10] suggested that plastic deformation effectively superimposed a reverse stress field which reduced the observed surface compression.

One other source of loss in the observed surface compression magnitude is due to the differential thermal contraction of surface layers upon cooling from the ion exchange temperature to ambient. Tyagi and Varshneya [11] have shown that the K-exchanged surface layers appear to have significantly higher thermal contraction coefficient relative to the substrate, accounting for a loss of ~20–25% in the magnitude of the surface compression. While this decrease reduces the disparity, the observed stress remains about 1/3rd that expected from as-melted molar volumes.

Laboratory measurements of stress in flat specimens do not yield useful information on stress buildup and relaxation for ion exchange times less than about 10 min, primarily because penetration of ions less than ~3 μm does not yield accurate stress measurements using any of the optical birefringence based techniques.

The use of molecular dynamic (MD) simulations elucidates some of the missing early time information, up to $\sim 10^{-7}$ s, with one important caveat. At present, MD simulations have not been attempted for glass geometries used in laboratory ion exchange experiments (uniaxial interdiffusion). The best one can do is to conduct the simulation on a cube of homogenized material where a fraction of alkali ions (0 to 100%) are swapped homogeneously and equilibrated under either NVT (constant number, volume, and temperature) constraint or under NPT (constant number, pressure, and temperature) constraint. Under NVT conditions, the pressure buildup from the stuffing of large ions and its subsequent relaxation, if any, are monitored. During NPT conditions (usually at $P = 0$), the volume change is monitored with time.

MD simulations at NVT in a 3-d ensemble yield pressure p that generates after uniformly swapping ions. To compare the results of the MD

simulation with those of ion exchange experiments involving thick plates having biaxial system of stresses, one needs to compute the equivalent plane-strain biaxial σ_{yy} (or thin slice plane stress σ_1) using the same strain generating mechanism. As shown in Appendix A, this is achieved for isotropic solids such as glass by recognizing

$$p = K \Delta V/V = E \Delta V/3(1-2\nu)V \quad (5)$$

where K is the bulk modulus, and ΔV is the change in the volume. Hence, to obtain biaxial stress σ_{yy} equivalent of $-p$, we multiply $-p$ by $[(1-2\nu)/(1-\nu)]$. Likewise, to obtain a thin slice equivalent stress σ_1 , we multiply $-p$ by $(1-2\nu)$. [Henceforth, the negative sign for compression will be dropped].

In the laboratory, only a few microns of the surface are usually exchanged. Biaxial stress is generated as a consequence of the constraining substrate. If we assume that the substrate is much larger in thickness than the surface exchanged region, then the biaxial stress on the free surface in the laboratory experiment could be compared to the biaxial equivalent of the pressure generated in the swapped volume in MD simulations using the above relations. Likewise, reported thin slice stress measurements can be scaled appropriately.

The purpose of this paper is to take an inventory of the buildup of surface compression at “zero time” and to reconcile the measured stresses with volume changes and relaxation estimates. We have conducted MD simulations for $15\text{Na}_2\text{O} \cdot 10\text{CaO} \cdot 75\text{SiO}_2$ (mol%) glass beginning with the instant of swapping the Na^+ ions with K^+ ions. Whereas most simulations in the past have only been carried out at the most to 3 or 4 ns, we have carried these simulations out from picoseconds to as much as 60 ns segments under cascaded NPT, NVT and NPT constraints. Next, we have carried out laboratory chemical strengthening experiments in a commercial soda lime silicate glass to determine the stress relaxation data in an effort to obtain zero time surface compression by extrapolation. Finally, the MD early time information and the laboratory later time data are pieced together to follow stress buildup at $t = 0$ and the subsequent stress relaxation and associated volume changes.

2. MD simulations

MD simulations used 6000 atoms of composition $\text{Na}_{620}\text{Ca}_{207}\text{Si}_{1552}\text{O}_{3621}$ which is close to $15\text{Na}_2\text{O} \cdot 10\text{CaO} \cdot 75\text{SiO}_2$ (mol%) glass with cubic periodic boundary conditions. Details of the procedure are given by Kreski et al. [12]

After cooling from 6000 K, the structure was equilibrated at 723 K (typical ion exchange temperature) for 20 ns under NPT ensemble. Next, the system was equilibrated under NVT ensemble for 10 ns to ensure the pressure development in the subsequent ion swapping step was not an artifact of the change of ensemble. After NVT equilibration, all Na ions were relabeled as K ions while maintaining NVT constraint. Pressure was monitored for as much as 60 ns for the 723 K simulation. The early stage relaxation under NVT after swapping was sampled at fractions of picosecond intervals for improved resolution. The constraint was then changed over to NPT ($P = 0$) and the volume expansion was monitored for 20 ns. Separately, the K-swapped random ensemble was equilibrated at 6000 K in the usual way and equilibrated under NPT conditions at 723 K to yield the CEAM structure.

3. Experimental procedure

The nominal chemical composition of the soda lime silicate glass chosen for the laboratory portion of this study was $12\text{Na}_2\text{O} \cdot 6\text{MgO} \cdot 9\text{CaO} \cdot 71\text{SiO}_2 \cdot 2(\text{Others})$ in mol%. Its relevant physical properties were strain point 511 °C, annealing point 554 °C, density 2.5 g/cm³ and stress optic coefficient 2.76 TPa⁻¹. Specimens, 25 mm $\times$ 25 mm $\times$ 0.4 mm thick were obtained from a commercial supplier. These specimens were ion exchanged in a bath of technical

KNO₃ (>99.6% pure) cumulatively in sequential steps of immersion time. The specimens were pulled out after 4 h of ion exchange, washed and dried. After cooling to room temperature (“RT”), their surface compression (“CS”) and case-depth (“DOL” or depth of layer) were measured using an Orihara FSM-6000LE ellipsometer [13]. The same specimens were reintroduced into the bath for the next incremental ion exchange, and so on. Since the specimens were thin, it was possible to introduce them into and withdraw them out of the molten salt bath rapidly without significant risk of thermal shock fracture.

4. Results of MD simulations

MD simulations (Fig. 1) show that, an NPT ($P = 0$) cube of the SLS glass has a molar volume of 23.92 cm³ (an ensemble volume of 82,200 Å³) at 723 K. This is represented as the host or “blue volume”

in Fig. 2. After swapping 100% with potassium ions under NVT at $t = 30$ ns, the instantaneous pressure at 723 K rises to ~6 GPa (Fig. 1b). Over the next 0.2 ps, the pressure rapidly decays to 3.4 GPa and further to 2.8 GPa in 4 ps. The rate of relaxation apparently slows down over the next few nanoseconds; pressure drops and appears to settle at ~2.1 GPa over 60 ns. When the ensemble is switched to NPT (Fig. 1a), the volume rises instantaneously to 25.06 cm³ (ensemble volume 86,100 Å³) and then asymptotically approaches ~25.24 cm³ (ensemble volume 86,700 Å³) over the next 20 ns; the instantaneous expansion being roughly 85% of the total change. This asymptotic volume is represented as the stuffed or “orange volume” in Fig. 2. The pressure-relaxed volume of 25.24 cm³ is still lower by about 50% of the change to the CEAM volume of ~26.4 cm³ (ensemble volume 90,700 Å³) represented as the “green volume” (Reader please note: For ease of identification, these states will be referred to by their color.).

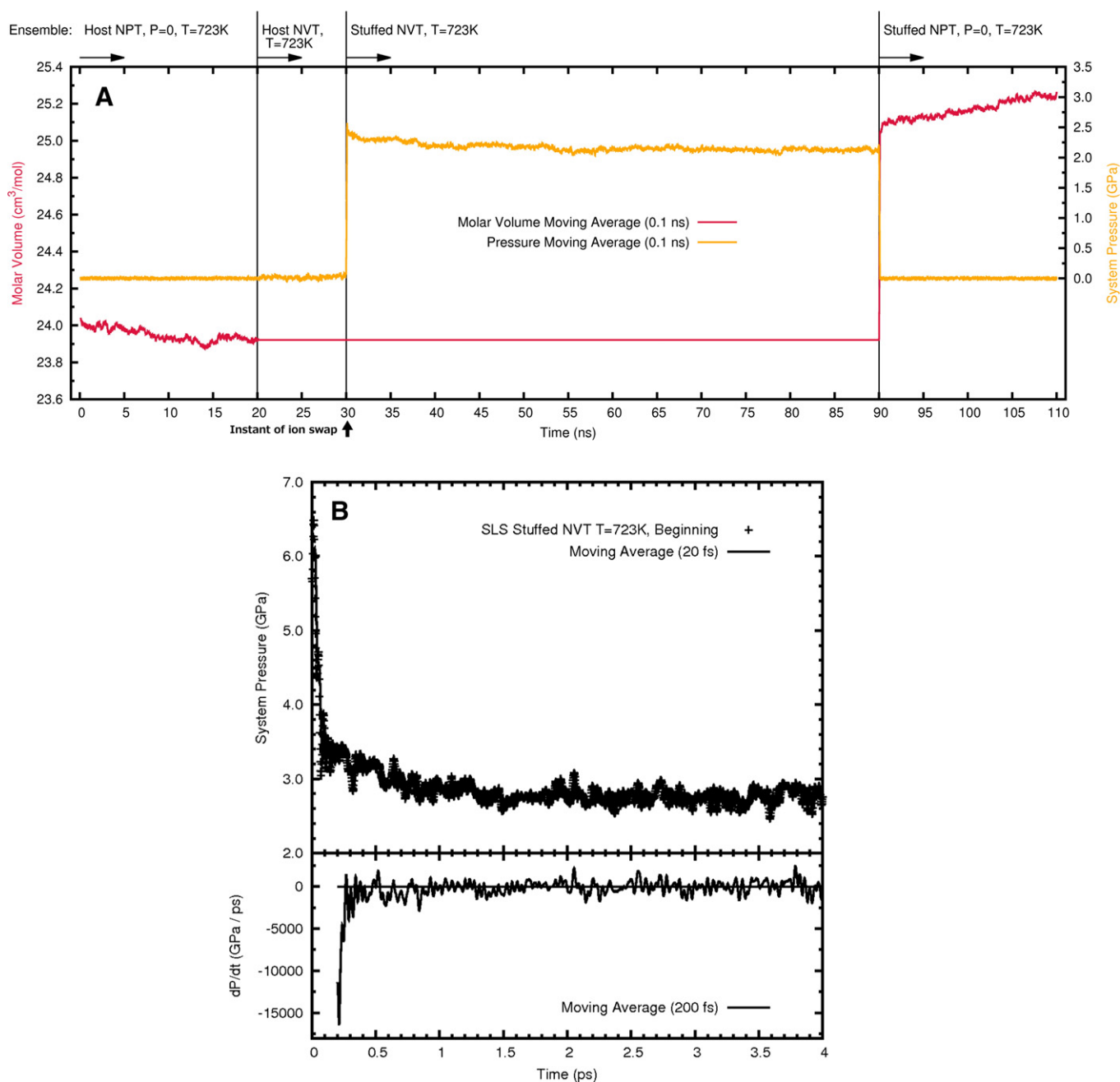


Fig. 1. a: Molecular dynamics simulation of a cube at 723 K and various boundary conditions of NPT and NVT. b: Details of the NVT simulation from the instant of swapping to a few picoseconds.

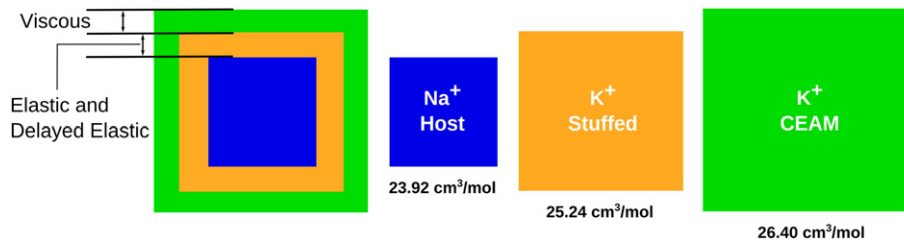


Fig. 2. Relationship between host, ion-stuffed, and CEAM glass (not to scale).

The volume magnitudes are similar to those reported earlier by Kreski et al. [12] for sodium silicate glasses. Some disparity with Tandia et al. [14–16] is ascribable to their lack of agreement with Bansal & Doremus data [17].

For the glass studied, the MD data can be summarized for 723 K as shown in Table 1.

5. Results of laboratory experiments

Biaxial surface compression $[\sigma_{yy}]_0$ measured at room temperature is plotted as a function of time in Fig. 3. Case-depth (DOL) varied between 8 μm and 50 μm and is not presented here. Initial examination revealed that the stress relaxation data could not be fitted to a single relaxation within the measurement time period, hence a stretched-exponential relationship of the type $[\sigma_{yy}]_{0,t} = [\sigma_{yy}]_{0,o} [\exp - (t/\tau)^\gamma]$, where τ = relaxation time and γ accounts for the non-Maxwellian behavior. Fig. 4 is a plot of the same data to show stretched-exponential fit to $\gamma = 3/7$ (the value usually quoted for structural relaxation, see ref. [18]). Values of τ and $[\sigma_{yy}]_{0,o}$, the extrapolated value of CS at time = 0, is computed over the scale of laboratory time and is shown in Table 2. $[\sigma_{yy}]_{0,o}$ (when measured at room temperature) varies between 720–800 MPa with relaxation times τ of the order of 1.4×10^5 to 2×10^6 s (~40 to 540 h) (Note: Because of limited decades on τ , our data could also be fitted to $\gamma = 3/5$, the value usually quoted for stress relaxation; it yields only slightly different values of $[\sigma_{yy}]_{0,o}$).

6. Discussion

Based on our simulation results and the simulation results reported by other investigators [12,14–16], we postulate that, following the initial swap of a small host by a larger invading ion, the total response of the system can be thought of as a composite of five stages:

- i) Initial energy jump: Instantaneous increase in energy at constant volume due to different interacting potentials. The energy spike generates a pressure spike of ~6 GPa at constant volume.
- ii) Local energy minimization: Over a few ps, the system slides down the energy hill to the nearest local energy minimum. During this

step, the Voronoi polyhedra about the foreign ions deform by increasing the K–NBO distances and decreasing the K–BO distances while keeping the foreign alkali ions in the cages of the host glass. There is no change in the topology of the Si–O network.

- iii) Fast “ β ” delayed-elastic (reversible) stress relaxation: During this step the alkali ions jump from one alkali ion cage into another during a time period from a few ps to a few ns. No change of Si–O network topology takes place during this step.
- iv) Slow irreversible “ α ” stress relaxation: During this step, which occurs after several ns, changes in the Si–O network topology (i.e., the short range structure of the glass) begin to take place. Processes such as CIRON (“cation-reduced relaxation of network”) suggested by Ingram et al. [19] may be some of the topology changing mechanisms. The slow relaxation could last several hours or days. At the end of this step all stresses are relaxed, however the structure of the glass is not in equilibrium (i.e., not the same as that of the CEAM glass). This we call the K^+ -stuffed or the “orange” state (Fig. 2). The stress-free ion-exchanged glass (orange volume) is a state not ordinarily arrived at on the V–T diagram by mere application of temperature, hence, termed a “forbidden” state by Mauro and Loucks [20].
- v) Structural relaxation at zero stress: During this slow step which may last years, the network undergoes long range cooperative structural relaxation (i.e., α -relaxation but at zero stress) reaching ultimately the final equilibrium structure corresponding to that of the CEAM glass (the green volume, Fig. 2).

Reader may kindly note that, in invoking the “fast β -relaxation” and “slow α -relaxation” above, we have incorporated the terminology and descriptions used by Binder and Kob [21]. Likely, there is considerable overlap between the stages. The relaxation of pressure with time scales of picoseconds, nanoseconds and perhaps microseconds and seconds must involve both the dilational deformation and flow (volume change with constant shape) driven by the dilational component of the stress tensor which may exceed the hydrostatic yield strength of the network and shear deformation and flow (shape change with constant volume) driven by the deviatoric component of the stress tensor associated with

Table 1
Summary of results from MD simulations.

Description	Pressure (GPa) at 723 K	$[\sigma_{yy}]_0$ (GPa) at 723 K	Ensemble volume ($\text{\AA}^3$)	Molar volume (cm^3/mol)
Host (SLS)	0	0	82,175	23.92
Stuffed (KLS), NVT	$t = 0^+$	5.5–6.5	4.0–5.0	23.92
	$t = 0.2$ ps	3.4	2.44	Same
	$t = 4.0$ ps	2.8	2.0	Same
	$t = 0.5$ ns	2.5	1.8	Same
	$t = 5$ ns	2.3	1.65	Same
	$t = 20$ ns	2.2	1.6	Same
	$t = 60$ ns	2.1	1.5	Same
Stuffed (KLS), NPT	$t = 0^+$	0	86,100	25.06
	$t = 20$ ns	0	86,710	25.24
CEAM (KLS)	0	0	90,697	26.40

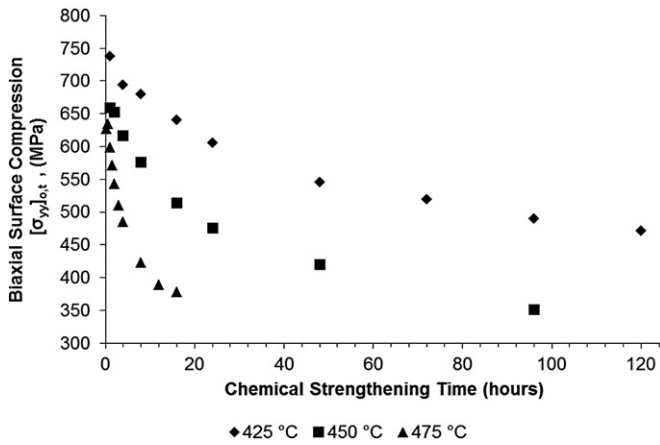


Fig. 3. Biaxial surface compression $[\sigma_{yy}]_{o,t}$ measured at room temperature as a function of time and ion exchange temperature.

the swelling of the alkali ion Voronoi polyhedra [22] which may exceed the shear yield strength of the network [1]. Deformations not reaching the yield strength are elastic or delayed elastic and, hence, reversible.

If we examine Fig. 1b, we may note that the initial pressure spike to ~6 GPa could be ascribed to the instantaneous response, Stage (i), of the network. At the end of this region (~0.2 ps) pressure drops to ~3.4 GPa. Densities of the host and the CEAM compositions at 450 °C from SciGlass v6.5 database [23] incorporating Priven-2000 method are 2.440 g/cm³ and 2.433 g/cm³. Corresponding molar volumes, hence, are 24.55 cm³ and 26.61 cm³, respectively. With the assumption that $K = 40$ GPa [23], the Cooper–Krohn expected pressure after ion exchange at 450 °C based on the as-melted end member molar volumes is 3.4 GPa which compares well with the MD observed pressure at 0.2 ps.

At the end of picosecond and nanosecond fast relaxations, Stages (ii) and (iii), begin to occur; the pressure settles to ~2.1 GPa which is equivalent to a biaxial compression $[\sigma_{yy}]_o$ value of ~1.5 GPa. If the ion exchange were stopped and the specimen cooled to the ambient, the stress would reduce ~20%, i.e., to 1.2 GPa due to the differential thermal contraction of surface layers [11]. Thus, we should expect a “zero time” room temperature measured experimental value σ_o of ~1.2 GPa.

On the other hand, if the ion exchange were allowed to continue (which is invariably the case), slow α -relaxation of the network, Stage (iv) and (v), associated generally with viscous flow commences. From the laboratory chemical strengthening data set, we note that the magnitude of σ_o as measured at RT is between 350 and 750 MPa. The σ_o values are clearly ion exchange treatment temperature-dependent. The

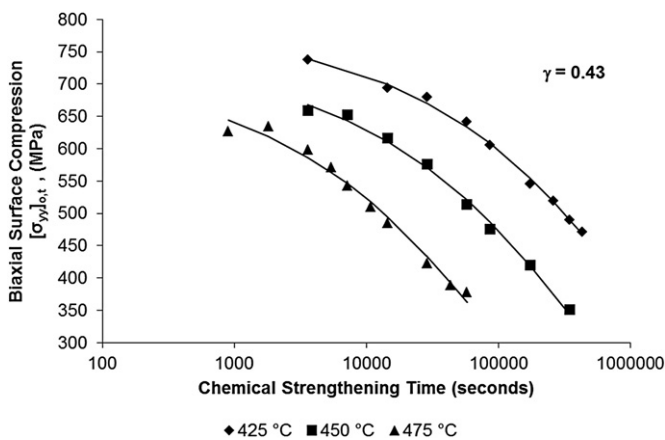


Fig. 4. Fit to stretched exponential of the type $[\sigma_{yy}]_{o,t} = [\sigma_{yy}]_{o,o} [\exp - (t/\tau)^\gamma]$, with $\gamma = 3/7$ for the time dependence of the surface compression in specimens ion-exchanged at three different temperatures.

Table 2

Estimated “zero time” biaxial stress $[\sigma_{yy}]_{o,o}$ (MPa) at room temperature and relaxation time constant τ (s) as a function of ion exchange temperature.

Ion exchange temp (°C)	$[\sigma_{yy}]_{o,o}$ (MPa)	τ (s)
425	790	1.94×10^6
450	744	6.3×10^5
475	723	1.37×10^5

stretched exponential fit over the experimental time range yields the “zero time” RT stress value of ~750 MPa at 723 K. Values of soda lime silicate glass shear viscosity at the temperature of ion exchange can be estimated to be $\sim 10^{16.5}$ Pa.s. If we assume the shear modulus to be ~30 GPa [23], then the relaxation time τ can be estimated to be $\sim 10^6$ s which is consistent with the values shown in Table 2. We conclude that stress decay upon continuing ion exchange treatment over experimental times is shear-flow related.

We can also follow the evolution of volume in the MD simulations. Host glass has a molar volume of 23.92 cm³ which is constrained (NVT) during the ion exchange. After undergoing relaxation for 60 ns under NVT, the pressure drops to 2.1 GPa. Upon releasing the volume constraint, volume increases to 25.24 cm³ within the next 20 ns which we label as the stress-free orange volume. System volume, however, falls ~50% short of its expected change during approach to the green volume (26.40 cm³). Transformation from the orange volume to the green volume can no longer be driven by stress, since the system is stress-free. Volume evolution must occur through an entropy-driven topological relaxation. Though the topology transforms, it is not a true first order thermodynamic transition and hence the orange state and the green state are not “polyamorphic” forms [24].

Buildup and relaxation of stress at various steps during ion exchange as a function of time (not to scale) can be schematically drawn as in Fig. 5. As stated above, if the ion exchange process were conducted for the traditional hours used and then cooled, one should expect a “zero time” biaxial compressive stress magnitude of ~1.2 GPa. Long time experiments will cause a stress-driven slow relaxation to zero stress (orange state) having topology different than the CEAM structure. From the stress-free orange volume, topology-driven relaxation to the CEAM structure could take years, but would occur more readily at elevated temperatures. Relaxation as observed in MD simulations and the laboratory experiments can be viewed on Log(time) scale as a composite, see Fig. 6. Two observations should be made:

- (1) There remains some disparity between the fraction of a second σ_o value of ~1.2 GPa (estimated room temperature) for simulations conducted at 723 K and that of the zero time prediction

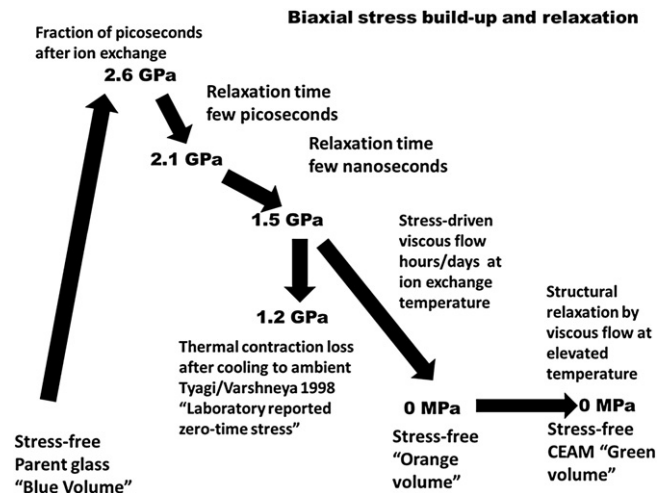


Fig. 5. Schematics of the buildup and relaxation of biaxial surface compression at various steps as a function of time (not to scale).

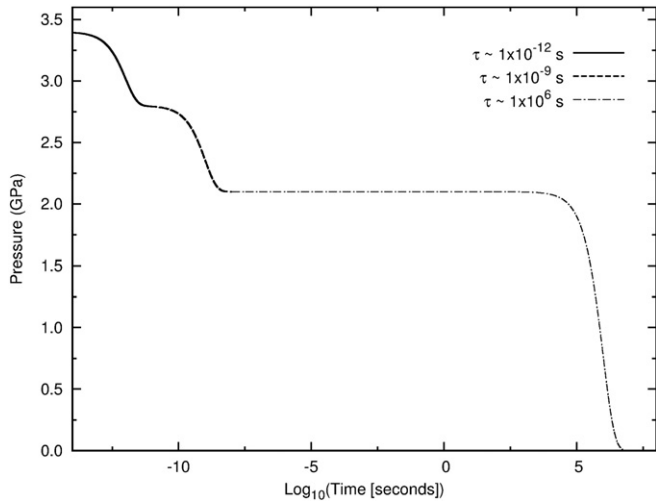


Fig. 6. Composite plot of stress relaxation as a function of Log(time).

magnitude ~ 750 MPa noted for 723 K exchange in Table 2 several minutes later. Clearly, we are missing relaxation information over the span of 10^{-9} s to 10^4 s (intermediate time constants), primarily due to time limitations associated with the MD simulations. The disparity is narrowed somewhat by considering Shaisha and Cooper's [25] single exchange experiments using electric field-assist ion exchange. A thin slice stress $[\sigma_y]_o$ of ~ 750 MPa (equivalent to 1 GPa biaxial stress $[\sigma_{yy}]_o$) was observed at room temperature by them in a soda lime glass tube after 300 s at 723 K. Shaisha and Cooper also noted that a compression of ~ 1 GPa did not relax much at 623 K over $100 \text{ s} < t < 7200 \text{ s}$ suggesting the possibility that all fast relaxations had already occurred within the first 100 s at 623 K. At 473 K, they were able to measure as high as 1.5 GPa after a 12,000 s exchange. It may be concluded that some of the non-shear viscosity related modes of relaxation that exist over seconds to a few minutes may well have been captured by Shaisha and Cooper. Our own experiments are currently underway to measure stress during the missing interval (early minutes) of ion exchange. Nonetheless, it may be suggested that the intrinsic value of the Cooper–Krohn theory in estimating stress development upon ion exchange in glass is limited at best.

- (2) The MD simulations box is only 6000 atoms-sized. It is likely that the relaxations in a real glass having many more atoms are smeared such that all the β -relaxations make up one stretched-exponential and the α -relaxations make up a second stretched-exponential as shown schematically in Fig. 6.

7. Conclusions

- (1) A comprehensive description of stress generation and relaxation has been developed using MD simulations for early time scale behavior and laboratory experiments for later time scale behavior.
- (2) A large portion of the discrepancy between Cooper–Krohn expectation of ~ 2.6 GPa surface compression (biaxial stress with plane-strain assumption) in a typical soda lime silicate glass at the exchange temperature and the experiment where only about 0.5–0.75 GPa stress develops is explained by “fast” (“ β ”) secondary structural relaxations where the time constants are picoseconds to nanoseconds.
- (3) The “fast” relaxations (“ β ”) secondary structural relaxations at the ion exchange temperature reduce the biaxial compression to ~ 1.5 GPa. Upon cooling to ambient, the differential shrinkage of surface layers with respect to the unexchanged bulk reduces the stress to ~ 1.2 GPa.

- (4) From the “zero time” value for typical laboratory measurements, the stress further relaxes to zero due to volume and shear viscous flow (primary “ α ” network relaxation) over minutes/hours/days of the experiment.
- (5) The ion-stuffed glass moves to a topologically different state from the as-melted structure and is not on the V–T ($P = 0$) diagram. This state is labeled “orange” state.
- (6) The stress-free “orange” state volume is $\sim 50\%$ short of the total expected change to thermodynamically stable equilibrium of the as-melted glass (CEAM volume) labeled “green” state.
- (7) The final relaxation from stress-free “orange” state to the stress-free CEAM “green” structure is a topology transformation. It occurs more readily at higher temperatures associated with volume and shear viscosities (primary “ α ” network relaxation) that are significantly reduced.
- (8) Some disparity remains between the ~ 1.2 GPa ending stress value in MD simulations and the laboratory observed ~ 0.5 to 0.75 GPa “zero time” values. It is possible that some intermediate time-scaled relaxation is yet to be studied. Nonetheless we are close enough to conclude that the “dilation anomaly” is largely resolved.
- (9) The use of the Cooper–Krohn theory for estimating surface compression generation after ion exchange in glass is limited at best.

Acknowledgments

The primary author (AKV) wishes to dedicate this paper to the memory of his late guru Professor Alfred R. Cooper for having made the initial (1969) and inadvertent error in estimating the linear network dilation coefficient; that the correctly calculated surface compression should have been a factor of 3 to 5 greater than those experimentally measured. That 1974 phone conversation triggered decades of long discussions on the “dilation anomaly”. Thanks are also due Professor Alastair Cormack of Alfred University and Professor Walter Kob of University of Montpellier for technical discussions, Robert McGinty of Mercer Engineering Research Center, Warner Robins GA for looking over Appendix A, and Ann Rossington for assistance with experimental measurements.

Appendix A

Application of the analogy to thermoelasticity allows the stress–strain relationships [26] to be written for a simple constrained cube which has been swapped completely by larger ions (“NVT” constraint) as:

$$\begin{aligned} e_{xx} &= \left(\frac{1}{E}\right) [\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})] + BC \\ e_{yy} &= \left(\frac{1}{E}\right) [\sigma_{yy} - \nu(\sigma_{xx} + \sigma_{zz})] + BC \\ e_{zz} &= \left(\frac{1}{E}\right) [\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})] + BC \\ e_{xy} &= \frac{\sigma_{xy}}{G}; \quad e_{yz} = \frac{\sigma_{yz}}{G}; \quad e_{xz} = \frac{\sigma_{xz}}{G} \end{aligned} \quad (A1-1)$$

where E is the Young's modulus and G is the shear modulus.

Equating $e_{xx} = e_{yy} = e_{zz} = 0$ and $\sigma_{xy} = \sigma_{yz} = \sigma_{xz} = 0$ for the constrained cube, we get

$$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma_{HYD} = -\frac{BEC}{1-2\nu} = -3KBC = -p \quad (A1-2)$$

where the bulk modulus $K = E/3(1 - 2\nu)$ and p is the internal pressure that develops due to the constraint.

The term $3BC = \Delta V/V_o$ = free volumetric expansion due to ion swapping when the cube is unconstrained.

For a biaxial plate of which the thickness is much smaller than the length and width dimensions and which has been completely ion exchanged in the $\pm x$ direction, we may assume that $\sigma_{xx} = 0$ and $\sigma_{yy} = \sigma_{zz}$. Also, if the plate is constrained at the edges as if it were being restrained by an unexchanged thick substrate, then $e_{yy} = e_{zz} = 0$. The plate is allowed to expand freely in the $\pm x$ direction. Substituting these boundary conditions in expression for e_{yy} in Eq. (A1-1) yields

$$\sigma_{yy} = \sigma_{zz} = -\frac{BEC}{1-\nu} \quad (\text{A1-3})$$

$$\text{And } e_{xx} = BC(1+\nu)/(1-\nu) \quad (\text{A1-4})$$

$$\text{Hence } \sigma_{yy}/(-p) = (1-2\nu)/(1-\nu) \quad (\text{A1-5})$$

Eq. (A1-5) shows that, when one of the edge constraints is released, the stress magnitude is expected to decrease relative to the internal pressure developed in a cube with “NVT” constraint. It may also be shown readily that the ratio of a hydrostatic stress $[-p = (1/3)(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})]$ to the suppressed component of the expected volumetric expansion for any situation is K as expected intuitively.

The biaxial stress on the surface can be measured non-destructively based on optical birefringence and surface refractometry. In prior literature, thin slices of the biaxial plate containing the diffusion direction were prepared and examined under polarized microscope for a direct measurement of the stress birefringence profile $[\sigma_y]_x$ as a function of the depth x which is a plane-stress value. In these specimens, $\sigma_{xx} = \sigma_{zz} = 0$, and it can be noted that the $[\sigma_y]_0 = -BEC = (1-\nu)[\sigma_{yy}]_0$.

References

- [1] A.K. Varshneya, Chemical strengthening of glass: lessons learned and yet to be learned, *Int. J. Appl. Glas. Sci.* 1 (2) (2010) 131–142.
- [2] O. Richmod, W.C. Lesli, H.A. Reidt, Theory of residual stresses due to chemical concentration gradients, *Trans. ASM* 57 (1964) (284–300).
- [3] A.R. Cooper, D.A. Krohn, Strengthening of glass fibers II: ion exchange, *J. Am. Ceram. Soc.* 52 (12) (1969) 665–669.
- [4] A.K. Varshneya, Influence of strain energy on kinetics of ion exchange in glass, *J. Am. Ceram. Soc.* 58 (3–4) (1975) 106–109.
- [5] ASTM, C1422–10, Standard specification for chemically strengthened flat glass, ASTM International, West Conshohocken PA, 2010.
- [6] A.Y. Sane, A.R. Cooper, Stress buildup and relaxation during ion exchange in glass, *J. Am. Ceram. Soc.* 70 (2) (1987) 86–89.
- [7] H.M. Garfinkel, C.B. King, Ion concentration and stress in a chemically tempered glass, *J. Am. Ceram. Soc.* 53 (2) (1970) 686–691.
- [8] A.K. Varshneya, Kinetics of ion exchange in glass, *J. Non-Cryst. Solids* 19 (1975) 355–365.
- [9] L.B. Glebov, V.G. Dokuchaev, N.B. Nikonov, G.T. Petrovskii, *Sov. J. Glas. Phys. Chem.* 14 (2) (1989) 131–137.
- [10] A.K. Varshneya, Physical properties of ion-exchanged and as-melted glass differ, *Glas. Res.* 10 (2/11(1)) (2001) 21–26, 51.
- [11] V. Tyagi, A.K. Varshneya, Measurement of progressive stress buildup during ion exchange in alkali aluminosilicate glass, *J. Non-Cryst. Solids* 238 (1998) 186–192.
- [12] P.K. Kreski, A.K. Varshneya, A.N. Cormack, Investigation of ion exchange ‘stuffed’ glass structures by molecular dynamics simulation, *J. Non-Cryst. Solids* 358 (2012) 3539–3545.
- [13] T. Kishii, Surface stress meters utilizing the optical waveguide effect of chemically tempered glass, *Opt. Lasers Eng.* 4 (1983) 25–38.
- [14] A. Tandia, K.D. Vargheese, J.C. Mauro, A.K. Varshneya, *J. Non-Cryst. Solids* 358 (2012) 316–320.
- [15] A. Tandia, K.D. Vargheese, J.C. Mauro, Elasticity of the ion stuffing in chemically strengthened glass, *J. Non-Cryst. Solids* 358 (2012) 1569–1574.
- [16] K.D. Vargheese, A. Tandia, J.C. Mauro, Molecular dynamics simulations of ion-exchanged glass, *J. Non-Cryst. Solids* 403 (2014) 107–112.
- [17] N.P. Bansal, R.H. Doremus, *Handbook of Glass Properties*, Academic Press, Orlando FL, 1986.
- [18] M. Potuzak, R.C. Welch, J.C. Mauro, Topological origin of stretched relaxation in glass, *J. Chem. Phys.* 135 (2011) (214502-1/7).
- [19] M.D. Ingram, J.E. Davidson, A.M. Coats, E.I. Kamitsos, J. Kapoutsis, Origins of anomalous mixed-alkali effects in ion-exchanged glasses, *Glastech. Ber. Glas. Sci. Technol.* 73 (4) (2000) 89–104.
- [20] J.C. Mauro, R.J. Loucks, Forbidden glasses and the failure of fictive temperature, *J. Non-Cryst. Solids* 355 (2009) 676–680.
- [21] K. Binder, W. Kob, *Glassy materials and disordered solids*, World Sci. (2011) 340ff.
- [22] A.K. Varshneya, The physics of chemical strengthening of glass: room for a new view, *J. Non-Cryst. Solids* 356 (2010) 2289–2294.
- [23] SciGlass Database v 6.5, ITC Inc. 2005. Newton MA.
- [24] O. Mishima, L.D. Calvert, E. Whalley, ‘Melting ice’ I at 77 K and 10 kbar: a new method of making amorphous solids, *Nature* 310 (1984) 393–395.
- [25] E. Shaisha, A.R. Cooper, Residual stress in singly and doubly ion-exchanged glass, *J. Am. Ceram. Soc.* 64 (1) (1981) 34–36.
- [26] B.A. Boley, J.H. Weiner, *Theory of Thermal Stresses*, 3rd printing, John Wiley & Sons, New York 1960, p. 245.